

Introduction to Blockchain

Encrypted Labs

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Who We Are

Encrypted Labs is a Blockchain consulting company focused on development and implementation of enterprise applications

- We are the first US company to partner with BigChainDB, a blockchain/database infrastructure that provides the scale, speed, and power to handle enterprise level requirements
- Our technical expertise with blockchain technology allows us to identify and design practical applications that solve customer problems and inefficiencies regardless of industry
- By creating in-depth, collaborative relationships with customers, we are able to create personalized solutions that are not offered by larger developers



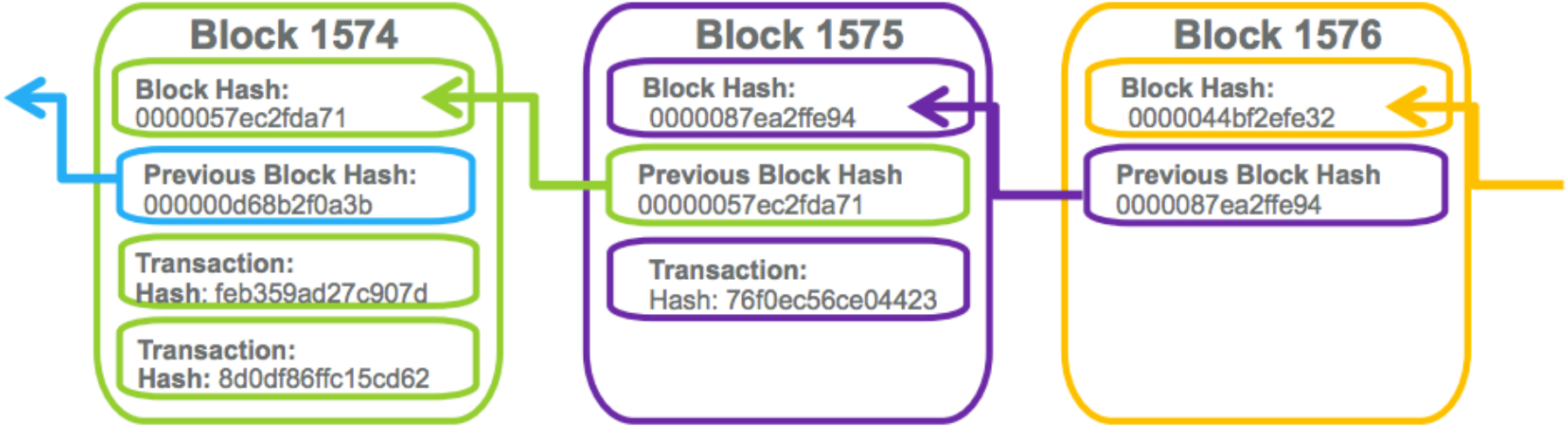
What is Blockchain?

Blockchain is a technology that revolutionizes the way that transactions are made between applications by establishing trust, accountability and transparency while streamlining business processes. Think of it as an OS for transactions that overcomes many traditional obstacles. It has the potential to greatly reduce the cost and complexity of getting things done by more efficiently organizing data and securing it.

- Blockchain was popularized through Bitcoin, but Blockchain itself is its own distinct underlying infrastructure onto which Bitcoin layered their own protocol.
- Blockchain opens the door to new interactions in business processes, across all industries that store data digitally



Blockchain Visualized

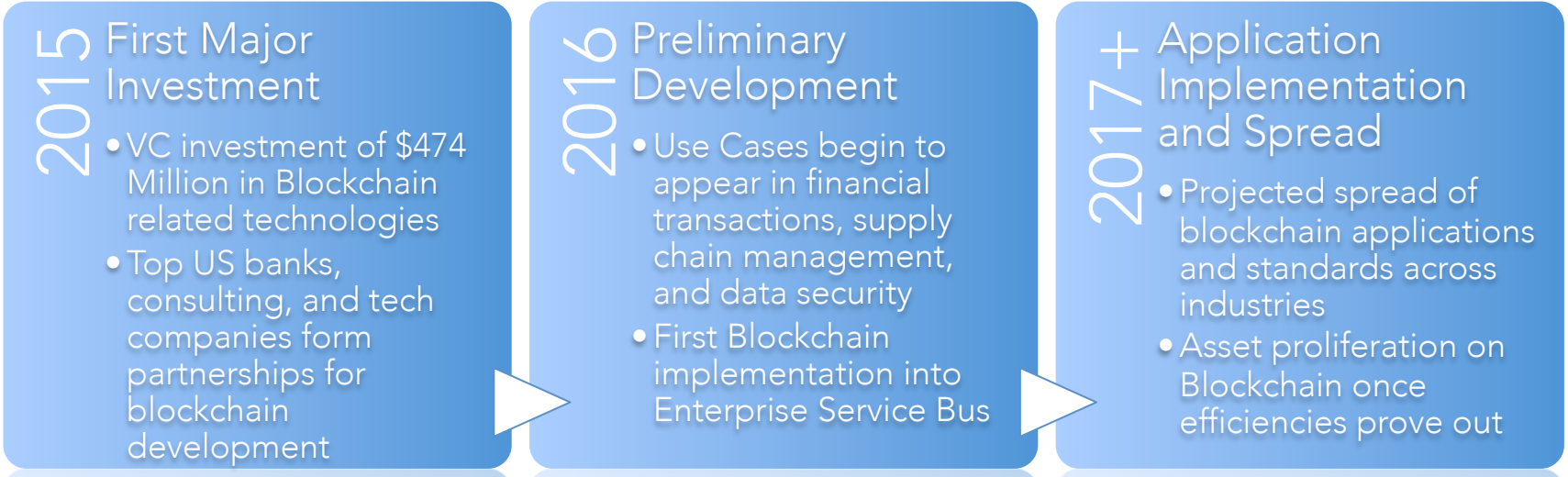


source: IBM - http://www.efinancelab.de/fileadmin/documents/results/video2016/20160704_Lang/01_Blockchain%20explained.pdf

Transactions are grouped into 'blocks', then stored forever in a 'chain' by linking each new block chronologically with the hash of the preceding block

Time Until Adoption

Blockchain is still considered an emerging technology, but many companies have already made significant investment into application development. Initiatives and use cases are proliferating at a rate that signals future adoption and industry-wide standardization.



Future of Blockchain

The attention that Blockchain has received in the past two years has created strong momentum for both software developers and beneficiary companies to take action towards creating useful applications.

- Once current Blockchain enterprise initiatives show signs of success, other companies will copy their success and we will see a new field of standardized, cross-industry Blockchain solutions
- Early adopters have an advantage over competitors because they have influence over Blockchain implementations, and benefit from efficiencies of Blockchain infrastructure long before slow-moving competitors



How Does Blockchain Affect Business?

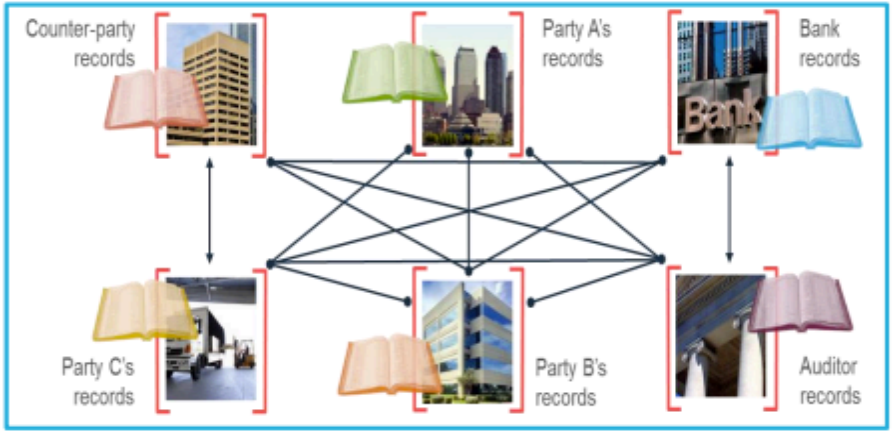
- Transparency, resilience, and cryptography protect data in the Blockchain
- Cryptography adds resilience to DBs to deter hackers
- Audit electronic actions to easily complete forensic analysis if a hack occurs
- Providing irrefutable evidence of electronic action
- Immutable logging and connection of data to create a tamper-proof chain of data
- Immutable data means accurate reporting and analytics
- These factors solve the inefficiencies of traditional transactional models



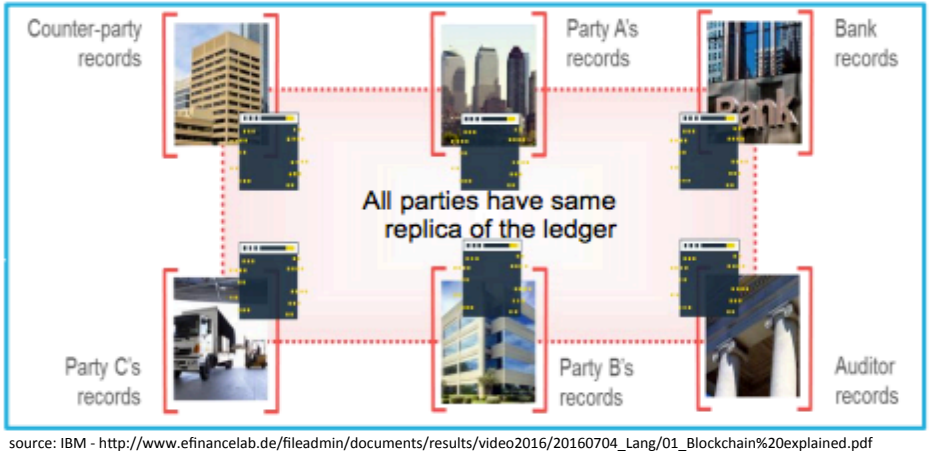
Before and After Blockchain

Financial Transactions Example

Without Blockchain



With Blockchain



source: IBM - http://www.efinancelab.de/fileadmin/documents/results/video2016/20160704_Lang/01_Blockchain%20explained.pdf

Inefficient | Expensive | Vulnerable

Consensus | Provenance | Immutability | Finality

A Blockchain Advantage - Immutability

Data is syndicated to all peers & signed by network of participants

Decentralized control is a federation of nodes with voting permissions that creates a super P2P network

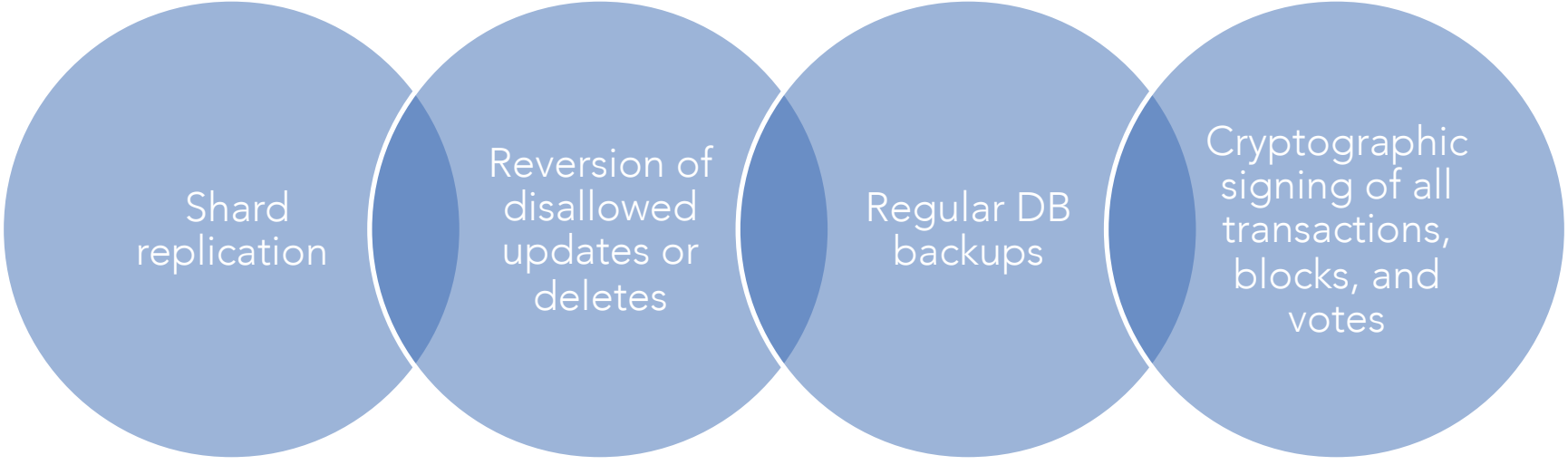
Each new block is chained to previous transaction & validated

If something was changed, the Blockchain would become invalid at that point and broadcast the error to all nodes

Immutability allows any fraud or error to be identified and corrected, eliminating any single point of failure and making it easy to have a correct record of all behavior at any point in time



Immutability Achieved



Current Initiatives

- Supply Chain: Fraud Prevention, leakage detection
- Cyber Security: Data Integrity – Hackers cannot change values
- Big Data: Auditing and Analytics
- Internet of Things: Registration, tracking usage data, payments
- Human Resources: Incentivizing employees, real-time performance reporting
- Financial: Settlements, post-trade data, back-office efficiencies
- Healthcare: Patient records management, security, reduced errors

Use Cases – Facing New Data Sabotage Threat

Background: The new trend in Cyber Attacks is not stealing data, but hackers changing data without the parent organization realizing some of their data has been corrupted.

- Public Sector - Cases of intelligence being altered so government officials act on untrue data/cannot trust what they are given.
- Private Sector - Cases of financials being altered so companies are making financial decisions incorrectly based on the outputs they are receiving from the data they have. Examples of tax records being changed, organ donor lists, security information, embezzlement.

How Blockchain Prevents Cyber Threats

Blockchain creates an immutable database where each entry is hashed in a block, and the current block has the hashes of all the previous blocks in it (distributed across nodes who all verify it), so if a past data entry is tampered with (and a majority of the nodes do not accept it) then it will make the block invalid, which both cancels the transaction and shows that data was tampered with.



Use Cases – Intellectual Property Tracking

BigchainDB can reduce licensing friction in channels connecting creators to audiences, and gives perfect provenance to digital artifacts.

- A typical music service has 38 million songs - BigchainDB could store this information in a heartbeat, along with licensing information about each song and information about use by subscribers.
- In another example, consider a medium-sized photo marketplace running 100,000 transactions a day; to put this on Bitcoin would cost $10,000 per day and tie up the Bitcoin network.



Use Case – Receipts and Certification

BigchainDB reduces legal friction by providing irrefutable evidence of an electronic action.

- BigchainDB is big enough that supporting information like receipts and certificates of authenticity (COAs) can be stored directly on it, rather than linking to the document or storing a hash.
- Legally-binding contracts can be stored directly on the BigchainDB next to the transaction, in a format that is readable by humans and computers.



Use Case – Supply Chain Tracking

By tracking high-volume physical assets along whole supply chain, BigchainDB can help reduce fraud, providing massive cost savings.

- Every RFID tag in existence could be entered on a BigchainDB.
- Smart contracts (decentralized processing), where the application must be fully decentralized and database functionality is crucial.
- Already being implemented to track diamonds and eliminate related fraud

Encrypted Labs Engagement Model



Summary

- Blockchain is coming & will transform many business operations
- Every corporate leader that wants their company to remain relevant knows about Blockchain and wants to learn more
- Encrypted Labs is leading & guiding companies through the learning, adoption, and implementation curves
- Early adopters will enjoy advantages relative to slow moving competitors, along with lower rates, and greater influence over direction of Blockchain tech



Relevant Links and Blockchain Publications

BigChainDB Whitepaper - https://www.bigchaindb.com/whitepaper/

Blockchain and non-financial use cases - http://www.mbnsolutions.com/blockchain-and-non-financial-services-use-cases/

Deloitte – New Blockchain Applications Will Accelerate Adoption - http://www.coindesk.com/deloitte-report-blockchain-use-cases-adoption/

UK Considering Government Applications for Blockchain Technology - https://bitcoinmagazine.com/articles/u-k-considering-government-applications-of-blockchain-technology-1462379719

17 Blockchain Disruptive Use Cases - https://everisnext.com/2016/05/31/17-blockchain-disruptive-use-cases/

Relevant Links and Blockchain Publications

Fast forward - **Rethinking enterprises, ecosystems and economies with blockchains**
http://www-01.ibm.com/common/ssi/cgi-bin/ssialias?htmlfid=GBE03757USEN#

UK Office of Science: **Distributed ledger technology: beyond block chain**
https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/492972/gs-16-1-distributed-ledger-technology.pdf

DTCC: **"Embracing Disruption – Tapping the Potential of Distributed Ledgers to Improve the Post-Trade Landscape,"**
http://www.dtcc.com/news/2016/january/25/new-dtcc-white-paper-calls-for-leveraging-distributed-ledger-technology

Oliver Wyman/Euroclear: **Blockchain in Capital Markets – The Prize and The Journey**
https://www.euroclear.com/dam/Brochures/BlockchainInCapitalMarkets-ThePrizeAndTheJourney.pdf



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